A Wireless Communication Gateway to Cross the Water-Ice-Air Boundary

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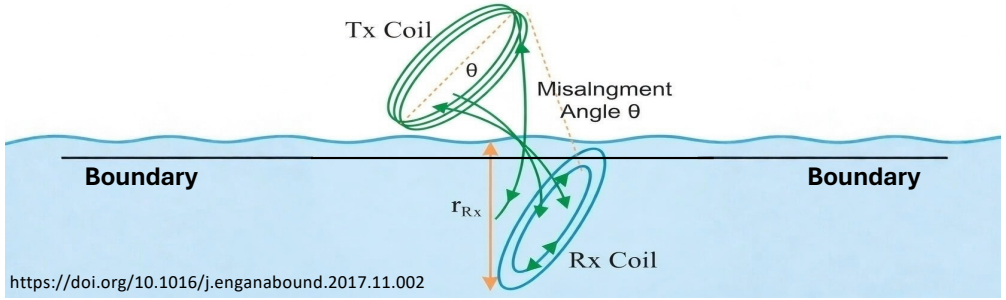
Relevance

1. The arctic serves as Earth's **atmospheric regulator** and contains the largest bank of **climate data** in any region.

2. A magnetic induction (MI) link offers reliable wireless **cross-medium communication** across the air-water boundary in **surface-obstructed** and harsh environments.

Challenges

1. Alignment Between Transmitter (Tx) & Receiver (Rx)
2. Conductivity Losses in Water



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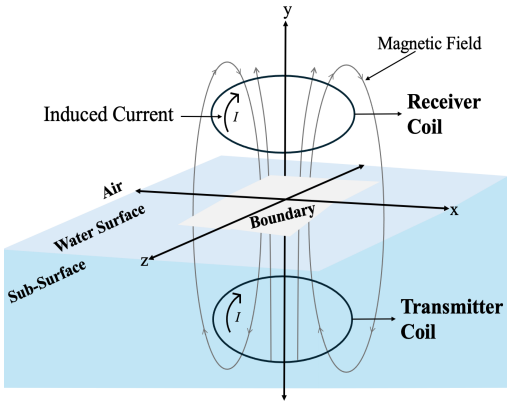
Methodology

1. Align Coils
2. Establish a Magnetic Field
3. Apply a Signal
4. Measure the Signal Strength
5. Calculate Path-Loss
6. Change Boundary Material



Future Work

Identify	Boundary Materials Similar to Ice
Design	The Magnetic Induction Link
Validate	Path-Loss Across the Boundary Layer
Conclude	Signal Strength as a Function of Boundary Thickness